12/02/24 AD

Hall Ticket Number:

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**MR-21**

**Code: B414CB4S**

**MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)**

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

**Object Oriented Programming Using Java**

*(Common to CSB, CSM & CSD)*

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

***Part-A (10 x 2 = 20 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 1. a) | List String comparison methods. | 2 | 1 | 1 | 1,2 |
| 1. b) | Mention the benefits of inheritance. | 2 | 1 | 1 | 1,2 |
| 1. c) | What is the datatype of the methods and variables in interfaces? | 2 | 1 | 2 | 1,2 |
| 1. d) | Define package. List any two predefined packages in java. | 2 | 1 | 2 | 1,2 |
| 1. e) | Specify different thread priorities. | 2 | 2 | 4 | 1,2 |
| 1. f) | Define generics. | 2 | 1 | 3 | 1,2 |
| 1. g) | Mention any two classes in java.util package. | 2 | 2 | 3 | 1,2 |
| 1. h) | What are the different ways to generate random number in Java? | 2 | 1 | 3 | 1,2 |
| 1. i) | List any four methods defined by Applet? | 2 | 1 | 5 | 1,2 |
| 1. j) | Differentiate between Panel and Window class. | 2 | 4 | 5 | 1,2 |

***Part-B (5 x 10 = 50 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 2. a) | Explain the Features of Java with examples. | 5 | 2 | 1 | 1,2 |
| 2. b) | List different types of access control and explain with suitable examples. | 5 | 2 | 1 | 1,2 |
| | **OR** | | | | |
| 2. c) | Define constructor and explain different types of constructors with examples. | 5 | 2 | 1 | 1,2 |
| 2. d) | Strings are immutable-Explain. | 5 | 2 | 1 | 1,2 |
| 3. a) | Write a java program to extend a superclass and implement at least two interfaces. | 5 | 3 | 2 | 1,2,3 |
| 3. b) | How does the Java run-time system know where to look for packages that you create? | 5 | 2 | 2 | 1,2 |
| | **OR** | | | | |
| 3. c) | If a class includes an interface but does not fully implement the methods defined by that interface, then that class must be declared as abstract-Justify. | 4 | 5 | 2 | 1,2 |
| 3. d) | Compare and contrast final, finally, finalize keywords. | 6 | 4 | 2 | 1,2 |
| 4. a) | Write a java program to create thread using Thread class. | 5 | 3 | 4 | 1,2,3 |
| 4. b) | Mention any 4 methods for console I/O. | 5 | 3 | 3 | 1,2 |
| | **OR** | | | | |
| 4. c) | Explain the commonly used methods for threads. | 4 | 2 | 4 | 1,2 |
| 4. d) | Differentiate how enum works within a class and switch statement. | 6 | 4 | 3 | 1,2 |
| 5. a) | List the advantages and limitations of Java iterators. | 5 | 1 | 3 | 1,2 |
| 5. b) | Analyze the different methods to read input of various data types using the Scanner class. | 5 | 4 | 3 | 1,2 |
| | **OR** | | | | |
| 5. c) | List and explain briefly three constructors in StringTokenizer class. | 5 | 2 | 3 | 1,2 |
| 5. d) | Explain four different types of Vector constructors. | 5 | 2 | 3 | 1,2 |
| 6. a) | Write a java program by passing parameters to applets. | 5 | 3 | 5 | 1,2,3 |
| 6. b) | Explain any two Layout Managers. | 5 | 2 | 5 | 1,2 |
| | **OR** | | | | |
| 6. c) | Write a java program to demonstrate key event handlers. | 5 | 3 | 5 | 1,2,3 |
| 6. d) | Explain Swing MVC components. | 5 | 2 | 5 | 1,2,3 |

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome